

Environment Modelling Lab Instructions

Objectives

- Set up a reproducible Python workspace on Windows using the provided `environment.yml`.
- Explore the raw point cloud and occupancy map data via the supplied Jupyter notebook.
- Build a baseline occupancy grid map with explicit ground removal using for example RANSAC and height thresholding.
- Experiment with an additional environment model of your choice (e.g., elevation map, stixels, signed distance field, mesh).

1. Setup (Windows + Anaconda)

1. Open an **Anaconda Prompt** and navigate to the cloned lab repository, e.g.

```
cd C:\path\to\lab_template
```

2. Create the project environment once:

```
conda env create -f environment.yml
```

The file defines an environment named `teaching`.

2. Notebook Walkthrough in VS Code

1. Open the project folder in **Visual Studio Code** (via `code .` or **File** → **Open Folder**).
2. Install the Python and Jupyter extensions (if prompted).
3. Use the VS Code command palette **Ctrl+Shift+P** → **Python: Select Interpreter** and pick the `teaching` environment so it appears as the active kernel.
4. Open `representations/raw_data_visualization.ipynb` inside VS Code and select the same `teaching` kernel in the top-right corner of the notebook.
5. Execute the notebook cells sequentially. The material demonstrates how to:
 - Load the provided point cloud and HD map data via `ad-env-utils`.
 - Visualise the raw data in 2D/3D projections.
 - Query map primitives needed for environment modelling.

3. Mandatory Task: Occupancy Grid Map

1. Design a ground-removal pipeline (e.g., height thresholding, plane fitting via RANSAC) to separate ground from obstacles in the point cloud.
2. Fuse the remaining points into an occupancy grid. Clearly document:
 - Grid resolution, spatial extent, and coordinate frame.
 - Sensor model / update rule (e.g., log-odds, binary).
 - Post-processing such as morphological filtering or inflation.
3. Validate the map visually.

4. Extension: Custom Environment Model

- Propose and prototype an additional model of the environment. Possible ideas:
 - Elevation or traversability map.
 - Signed distance field.
 - Mesh.
- Highlight the assumptions, data flow, and evaluation strategy. Document any limitations.